



Ordinary Differential Equations

Introduction to Differential Equations

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ENGR 213**

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Today's Lecture

- Initial-Value Problems
- Solution Curves without a Solution

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 1.2, 2.1**

Initial-Value Problems

- An **initial-value problem (IVP)** is a DE together with one or more conditions that specify the value of the unknown function, or its derivatives, at a particular value of the independent variable.

On some interval I containing x_0 , the problem:

$$\begin{cases} \text{solve: } \frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}) \\ \text{subject to: } y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1} \end{cases}$$

- where $y_0, y_1, \dots, y_{n-1}$ are arbitrarily specified real constants, is called an **initial-value problem (IVP)**.

- values of $y(x)$ and its first $n-1$ derivatives at a single point x_0 :
 $y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$, are called **initial conditions (IC)**.

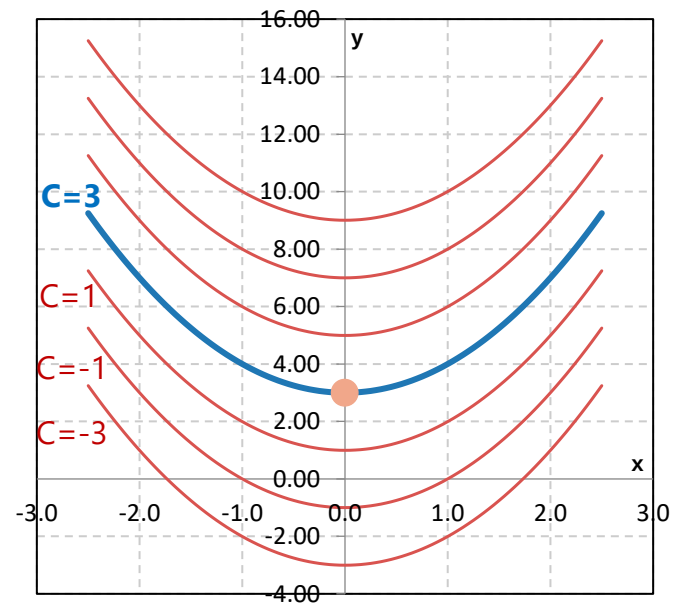
e.g.,: a first-order IVP

solve $\frac{dy}{dx} = 2x \longrightarrow y = x^2 + C$ * a family of solutions, one for every value of C

subject to: $y(0) = 3$ (initial conditions at $x=0$)

general solution: $y = x^2 + C \xrightarrow{\text{apply } y(0)=3} 3 = 0^2 + C \longrightarrow C = 3$

particular solution: $y = x^2 + 3$



e.g.,: a second-order IVP

solve $x'' + 16x = 0 \longrightarrow x = c_1 \cos(4t) + c_2 \sin(4t)$ * a two-parameter family of solutions

subject to: $x\left(\frac{\pi}{2}\right) = -2$ $x'\left(\frac{\pi}{2}\right) = 1$ (initial conditions at $x=\pi/2$)

general solution: $x = c_1 \cos(4t) + c_2 \sin(4t)$

$$\xrightarrow{\text{apply } x(\pi/2)=-2} -2 = c_1 \cos(2\pi) + c_2 \sin(2\pi) \longrightarrow c_1 = -2$$

$$x = -2\cos(4t) + c_2 \sin(4t) \quad \text{and} \quad x' = 8\sin(4t) + 4c_2 \cos(4t)$$

$$\xrightarrow{\text{apply } x'(\pi/2)=1} 1 = 8\sin(2\pi) + 4c_2 \cos(2\pi) \longrightarrow c_2 = \frac{1}{4}$$

$$\text{particular solution: } x = -2\cos(4t) + \frac{1}{4}\sin(4t)$$



□ Existence and Uniqueness

- Two fundamental questions arise in considering an IVP:
 - Does a solution of the problem exist?
 - If a solution exists, is it unique?

e.g., for a first-order IVP:

Existence $\left\{ \begin{array}{l} \text{Does } \frac{dy}{dx} = f(x, y) \text{ possess solutions?} \\ \text{Do any of the solution curves pass through the point } (x_0, y_0)? \end{array} \right.$

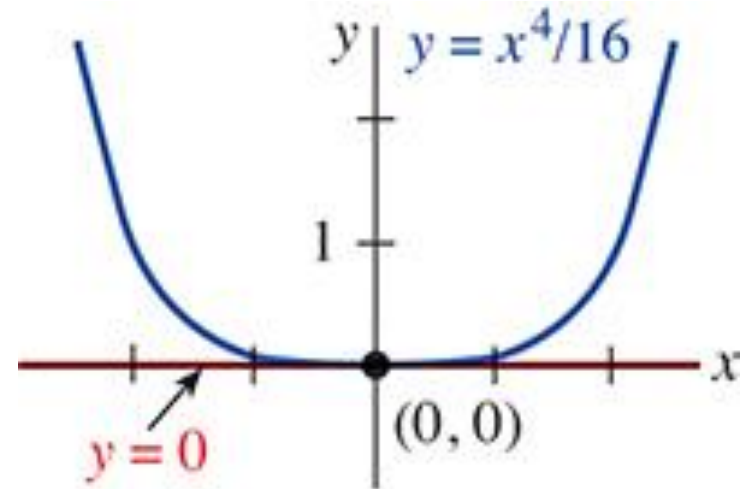
Uniqueness $\left\{ \begin{array}{l} \text{When can we be certain that there is precisely one solution} \\ \text{curve passing through the point } (x_0, y_0)? \end{array} \right.$

e.g.: $\frac{dy}{dx} = xy^{1/2}$ subject to: $y(0) = 0$

Two solutions: $y = \frac{1}{16}x^4$

and

$y = 0$



For a first-order initial-value problem: $\frac{dy}{dx} = f(x, y)$ $y(x_0) = y_0$

Theorem: Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$, $c \leq y \leq d$, that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f/\partial y$ are continuous on R , then there exists some interval $I_0 : (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$ defined on I_0 that is a solution of the first-order IVP.

Solution Curves without a Solution

- ❑ DEs can be studied in three different ways:
 - Analytically
 - Qualitatively
 - Numerically

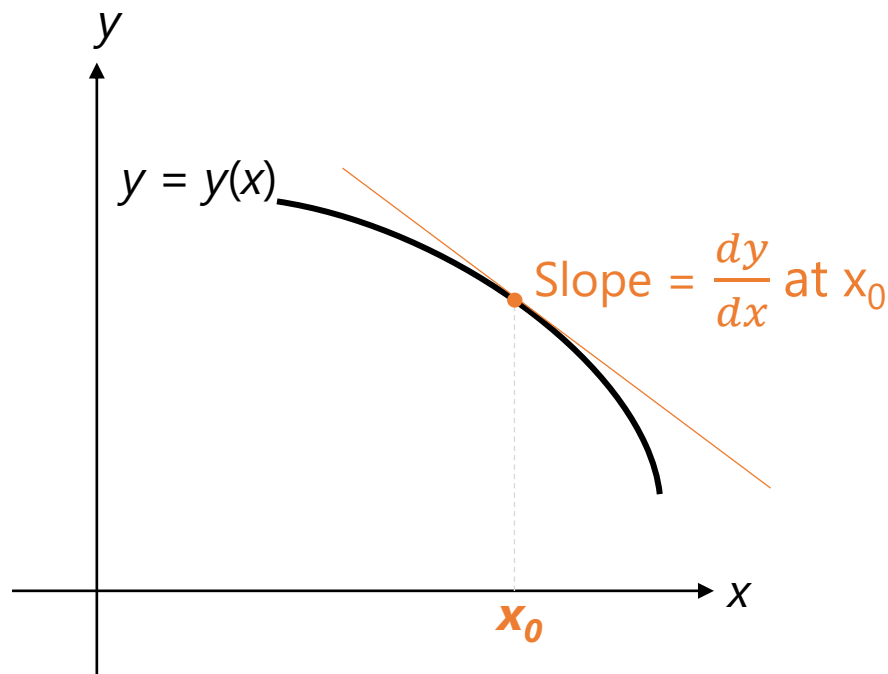
- ❑ DEs can be analyzed **qualitatively**, allowing us to approximate a solution curve without solving the problem.

- ❑ Two approaches are:
 - Direction fields
 - Autonomous first-order DEs

□ Direction Fields

▪ Slope

- Consider function $y = y(x)$ $\xrightarrow{\text{what is } \frac{dy}{dx} ?}$ $\frac{dy}{dx}$ gives slopes of tangent lines at points on the function graph.

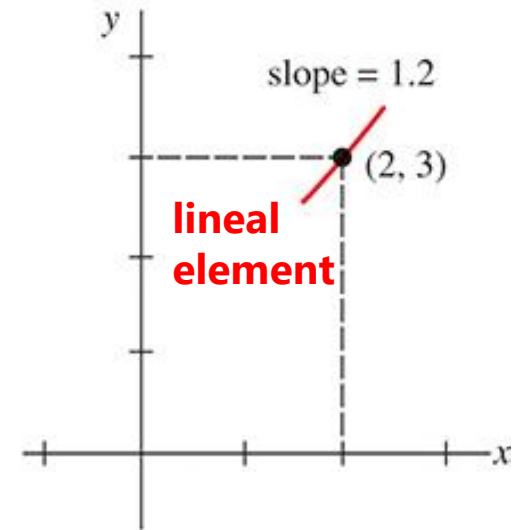


- Slope of the lineal element (tangent line) at $(x, y(x))$ on a solution curve is the value of dy/dx at this point.

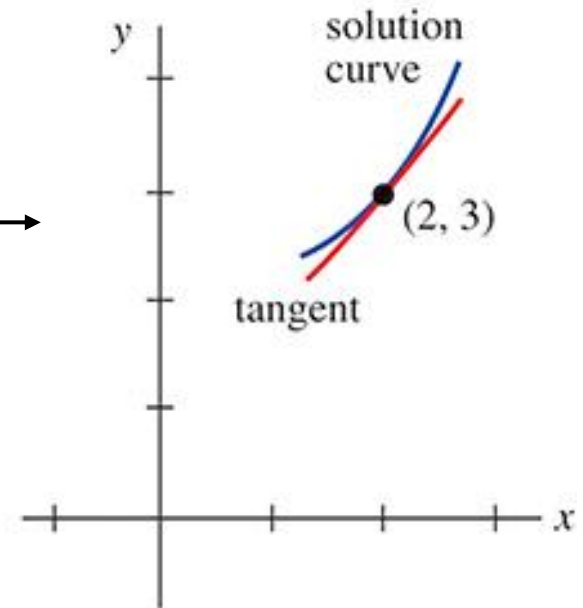
$$\frac{dy}{dx} = f(x, y)$$

e.g.: $\frac{dy}{dx} = \underbrace{0.2xy}_{f(x,y)}$

At the point $(x,y)=(2,3) \longrightarrow \frac{dy}{dx} = \overbrace{0.2(2)(3)}^{f(2,3)} = 1.2$



- So, any solution curve passing through (2,3) must have slope 1.2 at that point.

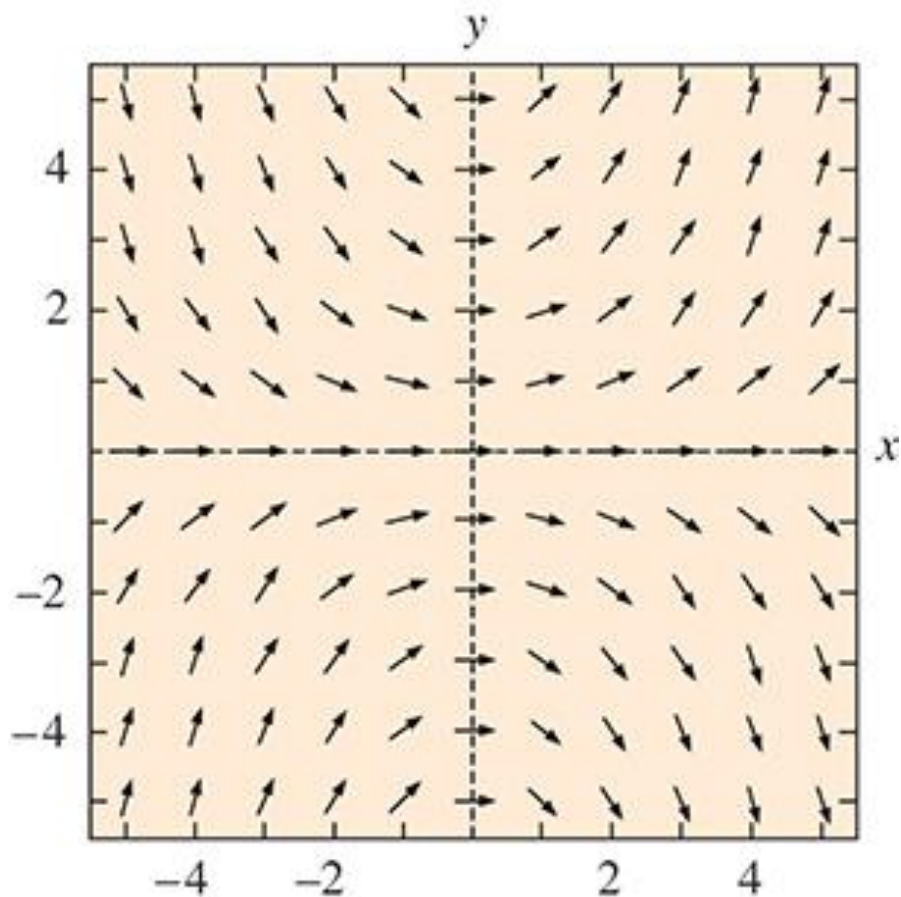


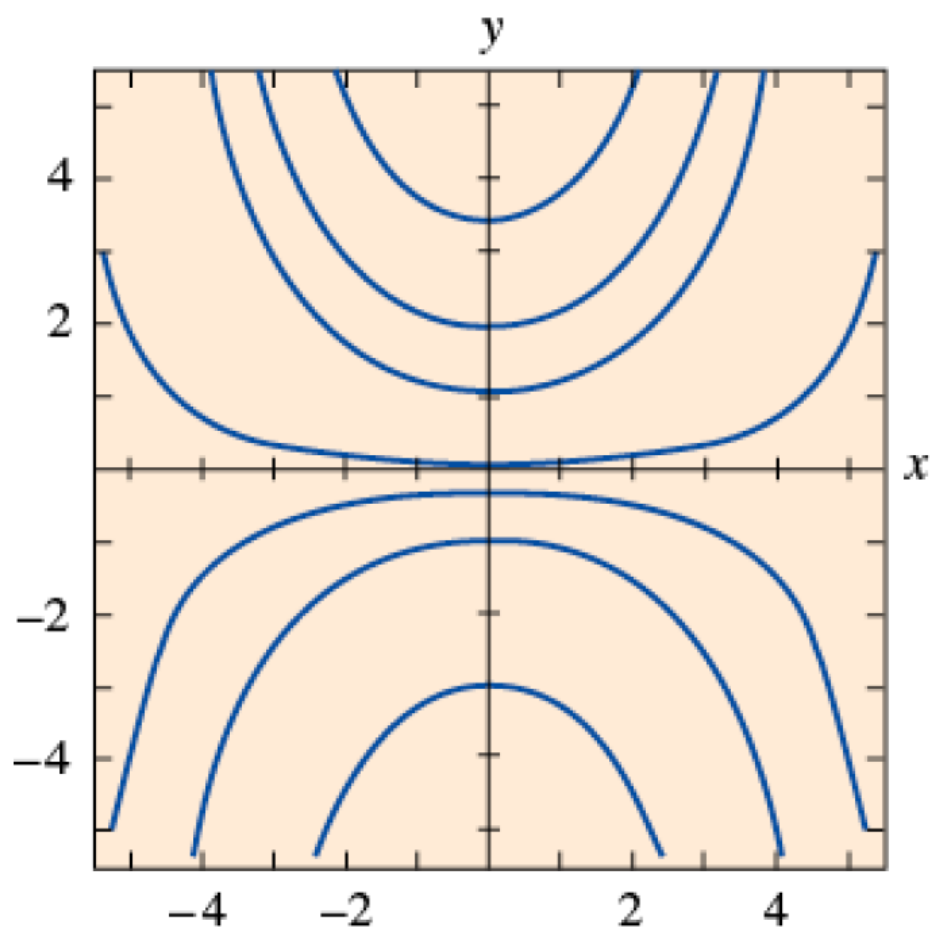
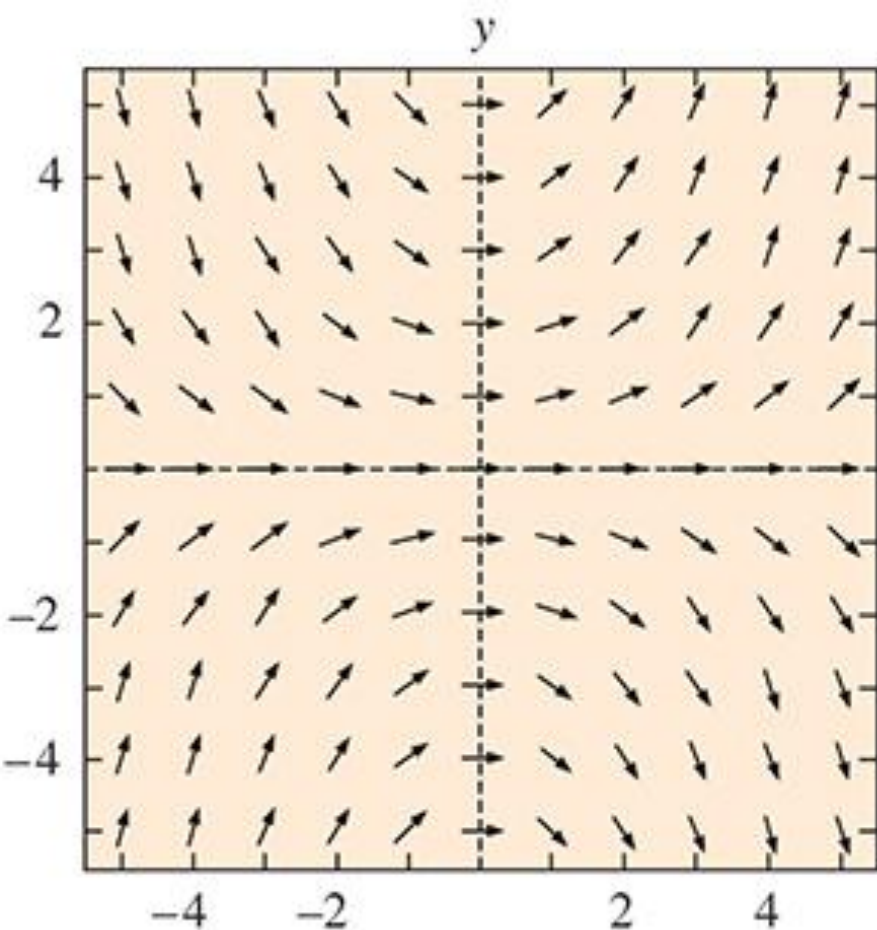
■ Direction Field

- If we evaluate $f(x,y)$ for many points and draw a lineal element at each point (x, y) , then the collection of all these lineal elements is called a **direction field** or a **slope field**.

e.g.: $\frac{dy}{dx} = 0.2xy$

(x,y)	dy/dx
$(-5,-5)$	5
$(-5,-4)$	4
...	...
$(1,-1)$	-0.2
...	...
...	...
$(5,5)$	5





general solutions: $y = ce^{0.1x^2}$

Direction Field Explorer: <https://c3d.libretexts.org/DirectionField/index.html>

□ Autonomous First-Order DEs

▪ DEs Free of the Independent Variable

- An ODE in which the independent variable does not appear explicitly is said to be autonomous. $\frac{dy}{dx} = f(y)$

e.g.: $\frac{dy}{dx} = 0.2xy$ $\swarrow f(x,y)$ $\frac{dy}{dx} = 1 + y^2$ $\swarrow f(y)$

▪ Critical points (equilibrium points)

- A real number c is a critical point of $\frac{dy}{dx} = f(y)$ if $f(c) = 0$.
- If c is a critical point of $\frac{dy}{dx} = f(y)$, then $\mathbf{y(x) = c}$ is a **constant solution** (or **equilibrium solution**) of the autonomous DE.

e.g.: $\frac{dy}{dx} = y(1 - y)$

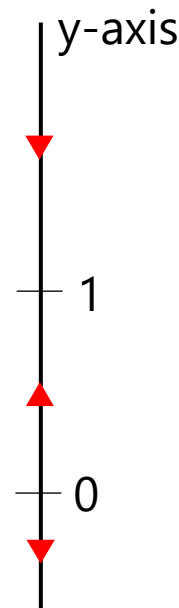
critical points

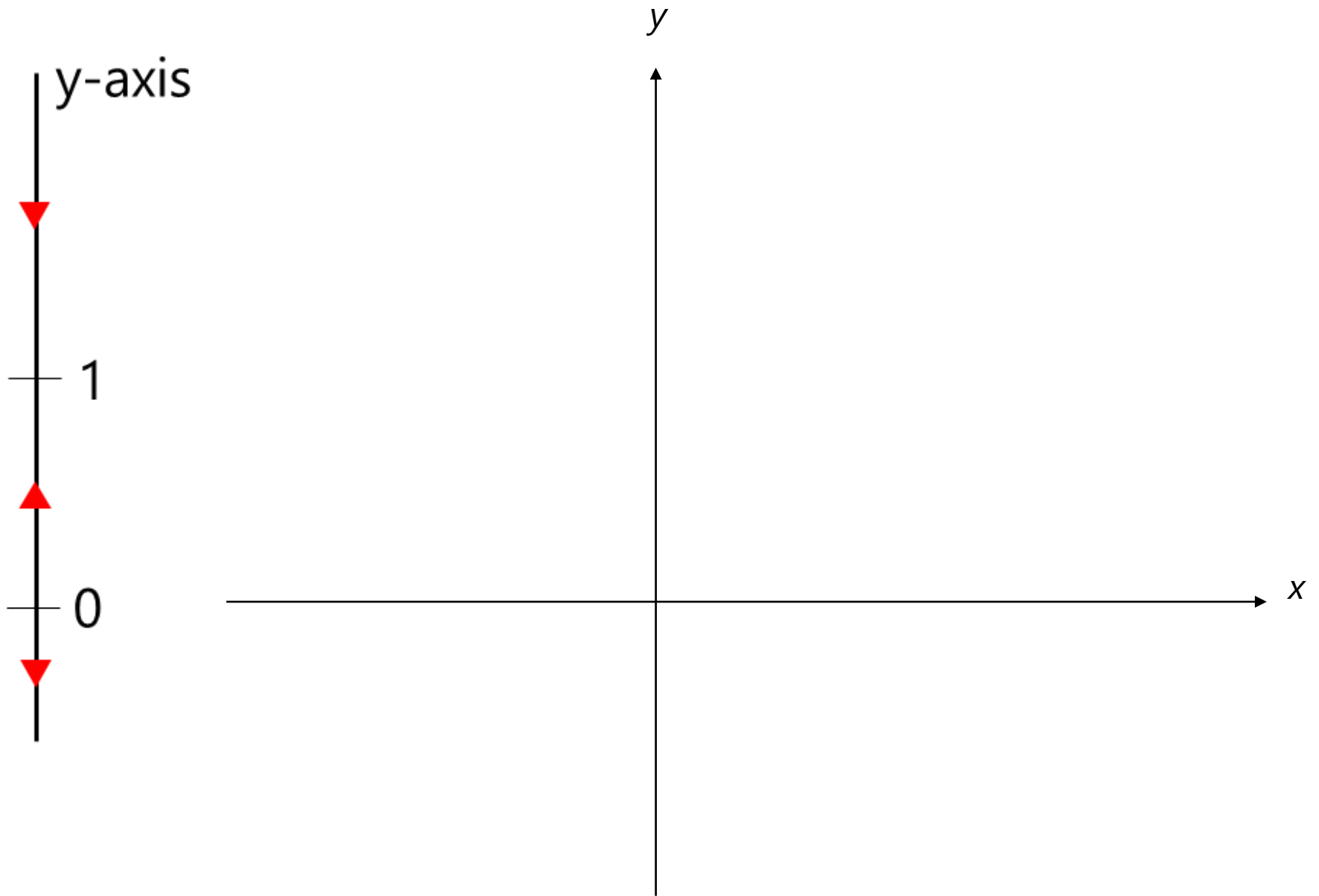
At critical point: $f(c) = 0 \longrightarrow f(y) = y(1 - y) = 0 \longrightarrow \begin{cases} y = 0 \\ y = 1 \end{cases}$

Constant solutions (or equilibrium solutions): $y(x) = 0, \quad y(x) = 1$

y interval	Sign of f(y)	y'(x)	solution y(x) behavior
$(-\infty, 0)$	-	< 0	decreasing ↓
$(0, 1)$	+	> 0	increasing ↑
$(1, \infty)$	-	< 0	decreasing ↓

- This is called a **one-dimensional phase portrait**, or **phase portrait**, of the DE.
- The arrows on the phase line show the direction of the non-constant solution.





- Classifying a critical point based on the arrows around it:

		Behavior	Classification
(a)	$\uparrow \bullet \downarrow$	solutions move toward c from both sides	Asymptotically stable - attractor
(b)	$\downarrow \bullet \uparrow$	solutions move away from c from both sides	Unstable - repeller
(c)	$\uparrow \bullet \uparrow$	toward from one side, away from the other	Semi-stable
(d)	$\downarrow \bullet \downarrow$	toward from one side, away from the other	Semi-stable

